

A Taste and Texture Experiment: Cookies Substituting Eggs With Applesauce, Yogurt, and Flaxseed

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2/8/2025

Abstract

In this experiment, a standard cookie recipe was used to discover which egg substitute out of apple sauce, greek yogurt, and flaxseed paste would prove most efficient taste and texture wise. The standard cookie recipe was used as the control, along with three separate batches, each one testing one of the substitute variables. The doughs were prepped and baked identically, with the egg substitutes being added in as a final ingredient for all, to minimize any other variability. Qualitative data was collected on the cookies from their moistness, taste, size, to texture. The results proved the applesauce and yogurt substitute cookies to be most similar to the control. This experiment proposes how easy it is to modify a standard baking recipe, which can prove to be helpful when missing ingredients or if needed for dietary restrictions.

Introduction

In baking cookies, it is common practice to use eggs, as they provide structure due to the protein components within their chemical makeup (Redpath Sugar, 2024). However, egg substituted treats are becoming all the rave, due to a large population's dietary restrictions. This experiment aims to test three egg substitutes; applesauce, unflavored greek yogurt, and flaxseed paste made through a combination of flaxseeds and water.

Applesauce proves a good substitute to eggs because of the water properties within its components (Gómez, 2021). As eggs contain about 75% water, applesauce contains about 88% water. While eggs contain protein and healthy fats, applesauce mostly contains fats and sugar. It proves to be a good substitute, however it will provide more moisture and cakiness to the cookie (Hayek, 2013). Yogurt provides a good substitute to eggs due to its large protein content. Greek yogurt contains about 4-6 grams of protein in a serving, whereas a small egg contains 4-7 grams

of protein. The range being almost identical proves to be a good substitute. Yogurt also contains polyunsaturated fats, allowing for the reactions within baking a cookie to proceed seamlessly (Hadjimbei, 2022). Finally, by combining ground flax seeds and water, flaxseed paste will appear in only a couple of minutes. While flax seeds on their own lack the water content needed to replace eggs, the gel provides a thickening agent for the cookies along with substituting the necessary proteins for structural support (Lorenc, 2022). Flaxseed gel provides a good means to egg substitution, although it does add more fibers than a standard recipe would.

References

- Gómez, M., & Martínez, M. M. (2021). Fruit and vegetable by-products as novel ingredients to improve the nutritional quality of baked goods. *Trends in Food Science & Technology*, 116, 118–132. <https://doi.org/10.1016/j.tifs.2021.07.006>
- Hadjimbei, E., Botsaris, G., & Chrysostomou, S. (2022). Beneficial effects of yoghurts and probiotic fermented milks and their functional food potential. *Foods*, 11(17), 2691. <https://doi.org/10.3390/foods11172691>
- Hayek, S. A., & Ibrahim, S. A. (2013). Consumer acceptability of chocolate chip cookies using applesauce as a fat (butter) substitute. *Emirates Journal of Food and Agriculture*, 25(3), 159–168. <https://doi.org/10.9755/ejfa.v25i3.10828>
- Lorenc, F., Jarošová, M., Bedrníček, J., Smetana, P., & Bárta, J. (2022). Structural characterization and functional properties of flaxseed hydrocolloids and their application. *Foods*, 11(15), 2304. <https://doi.org/10.3390/foods11152304>

Redpath Sugar. (2024.). *Science of cookies*. Retrieved February 8, 2025, from

<https://www.redpathsugar.com/recipe/science-cookies#:~:text=When%20you%20cream%20butter%20and,air%2C%20leading%20to%20denser%20cookie>